$$VLGn := \begin{bmatrix} 7199.56\angle 0 \\ 7199.56\angle -120 \\ 7199.56\angle 120 \end{bmatrix} \qquad U := \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} Iam \\ Ibm \\ Icm \end{bmatrix} := \begin{bmatrix} 249.97\angle -24.5 \\ 277.56\angle -145.8 \\ 305.54\angle 95.2 \end{bmatrix} \qquad j := \sqrt{-1}$$

$$A := U$$

$$Zabc := \begin{bmatrix} .8666 + j\cdot 2.0417 & .2955 + j\cdot .9502 & .2907 + j\cdot .7290 \\ .2955 + j\cdot .9502 & .8837 + j\cdot .9852 & .2992 + j\cdot .8023 \\ .2907 + j\cdot .7290 & .2992 + j\cdot .8023 & .8741 + j\cdot 2.0172 \end{bmatrix}$$

$$B := Zabc$$

$$\begin{bmatrix} Ian \\ Ibn \\ Icn \end{bmatrix} := \begin{bmatrix} Iam \\ Ibm \\ Icm \end{bmatrix} = \begin{bmatrix} 249.97\angle 0.633 \\ 277.56\angle -1.287 \\ 305.54\angle 0.952 \end{bmatrix}$$

$$VLGm := A\cdot VLGn - B\cdot \begin{bmatrix} Ian \\ Ibn \\ Icn \end{bmatrix} = \begin{bmatrix} 7.218\cdot 10^3\angle -0.102 \\ 7.291\cdot 10^3\angle -0.66 \\ 6.931\cdot 10^3\angle 0.518 \end{bmatrix}$$

$$Dv := \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ -1 & 0 & 1 \end{bmatrix}$$

$$Vllm := Dv\cdot VLGm = \begin{bmatrix} 3.998\cdot 10^3\angle 1.207 \\ 7.91\cdot 10^3\angle -1.604 \\ 4.327\cdot 10^3\angle 1.842 \end{bmatrix}$$

$$Vavg := 6916.20$$

$$dVi := Vavg - VLGm = \begin{bmatrix} 782.36\angle 1.915 \\ 4.621\cdot 10^3\angle 1.317 \\ 3.547\cdot 10^3\angle -1.316 \end{bmatrix}$$

$$Vunbalance := .4119$$

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$$\begin{bmatrix} Sa \\ Sb \\ Sc \end{bmatrix} := \begin{bmatrix} \overline{Iam} \cdot (7199.56\angle 0) \\ \overline{Ibm} \cdot (7199.56\angle -120) \\ \overline{Icm} \cdot (7199.56\angle 120) \end{bmatrix} \cdot \frac{1}{1000} = \begin{bmatrix} 1.451 \cdot 10^3 - 1.064i \cdot 10^3 \\ 1.57 \cdot 10^3 + 1.237i \cdot 10^3 \\ 2.079 \cdot 10^3 - 718.517i \end{bmatrix}$$

$$Tn := \begin{bmatrix} -.4292 - j \cdot .1291 & -.4476 - j \cdot .1373 & -.4373 - j \cdot .1327 \end{bmatrix}$$

$$In := Tn \cdot \begin{bmatrix} Iam \\ Ibm \\ Icm \end{bmatrix} = 215.45\angle -148.286^\circ$$

$$Ig := 32.5\angle -77.6$$

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